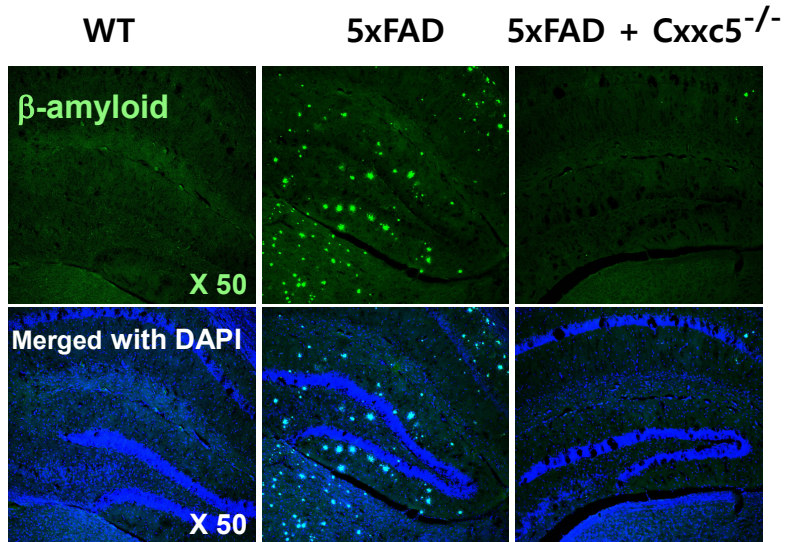


Alzheimer's Disease Program

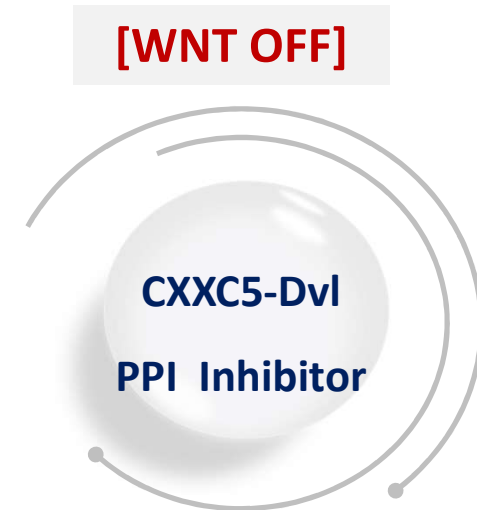


An Orally Available Small-Molecule Therapeutic Approach for Alzheimer's Disease

Targeting CXXC5, a Key Disease-Mediating Factor



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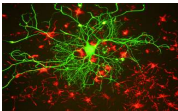
Recovery of Regenerative WNT signaling
by specific functional inhibition of
CXXC5, a root cause

WNT ON

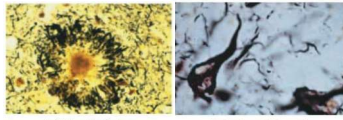
Alzheimer's Diseases (AD) and Unmet Needs/ Issues

- **Alzheimer's disease(AD):** Age-associated neurodegenerative disease (ND) characterized by progressive loss of **cognitive function**, memory and other associated **neural-inflammation** related issues. **Extracellular amyloid deposits** **intracellular neurofibrillary tangles** are the pathological hallmarks of the disease.

Neuronal-inflammation



Pathological plaques



Cognitive deficits



- Memory loss
- Poor judgment,
- Loss of spontaneity and sense of initiative.

Current Therapeutics; Amyloid, p-Tau, and GLP-1, NMDA Receptor Agonist...

Current Unmet Needs; Safe and convenient treatments that produce a large, durable clinical benefit; work in moderate-to-severe AD; address tau, inflammation, synaptic dysfunction and neuronal loss—not amyloid alone—and ideally promote functional neuronal repair or regeneration

No Regenerative
Disease-Modifying Therapy



Major driver ?



New Target!

Targeting intractable Alzheimer's Disease associated with impaired cellular regeneration by reactivating regenerative WNT signaling in damaged brain tissue



Root Cause Identification

We identified **CXXC5**, a negative regulator of WNT/ β -catenin signaling, as a key disease-driving/mediating factor that suppresses regenerative capacity in Alzheimer's disease (AD) patient tissues.

Pathological Implication: Cytosolic CXXC5 increment with WNT signaling suppression was observed all AD patient tissue cells.

Target Confirmation; When crossed with *Cxxc5* knock out mice, 5xFAD Alzheimer' disease model mice did not develop any pathological phenotype of AD.

Human Indication/ Genetic de-risking
(Target validation)



Small Molecule PPI Inhibitor Platform Targeting the Root-Cause Disease Driver

A novel screening platform to identify CXXC5-Dishevelled protein-protein interaction (PPI).

Restores regenerative WNT/ β -catenin signaling in patient tissue cells **by selective inhibiting the cytosolic function of CXXC5, a negative regulator**, through blockade of the CXXC5-Dishevelled PPI.

Platform Technology
(Innovation)



Disease-Modifying Therapy

Improvement of multiple pathological phenotypes of AD;

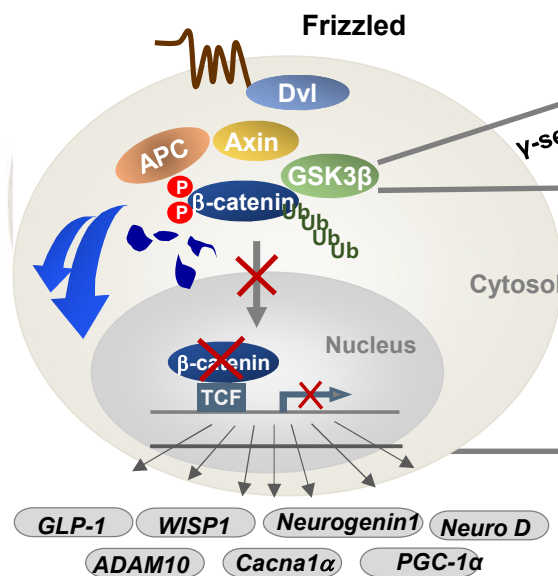
Reduction of amyloid pathology,
Recovery of neuronal function, &
3) Cognitive improvement

Pharmacologic Efficacies
(Translation)

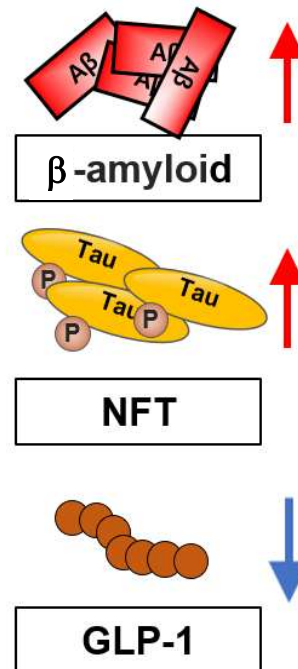


Major developmental targets for AD drugs are all related with WNT pathway

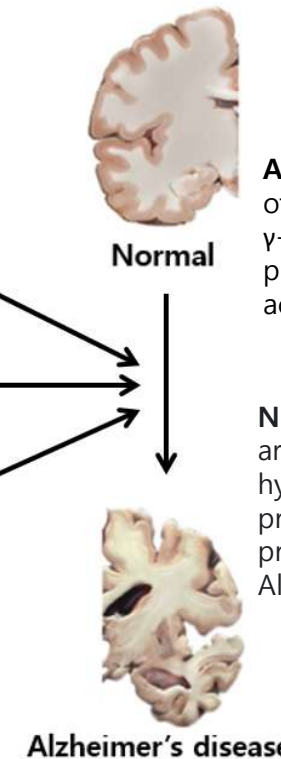
① Wnt target gene down regulation & GSK3 β activation



② Increase of β amyloid, Tau protein & Decrease of GLP-1



③ Alzheimer's disease



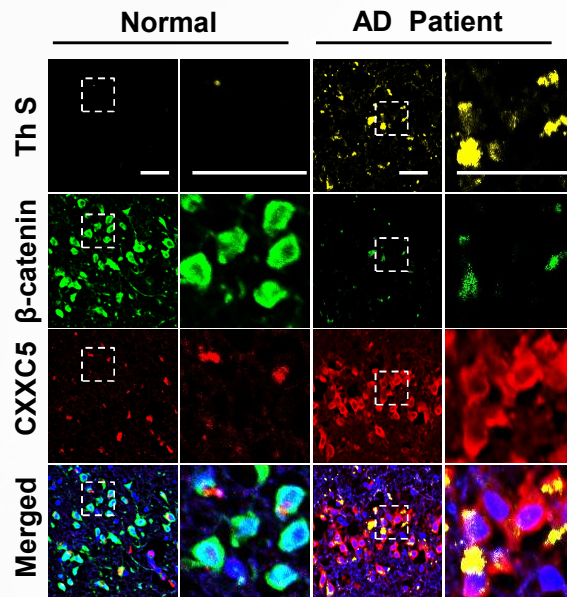
Amyloid β ; Down-regulation of Amyloid-beta by inhibiting γ -secretase complex phosphorylation via de-activation of GSK3 β .

Neurofibrillary tangles (NFTs) are aggregates of hyperphosphorylated tau protein; commonly known as a primary biomarker of Alzheimer's disease.

Overall Development

CXXC5 as a target for development of AD drug [Validation: Human, Genetic and Drug Efficacy]

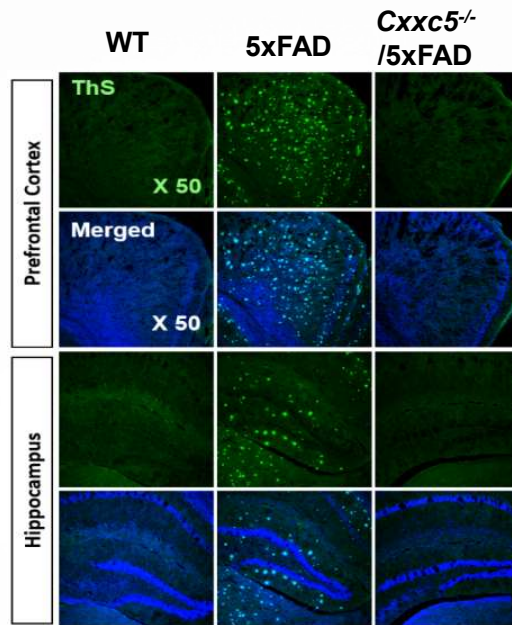
1) Target Indication By both patents and model animals



1

AD patients showed cytosolic CXXC5 overexpression with suppression of WNT/ β -catenin signaling in Hippocampus tissue

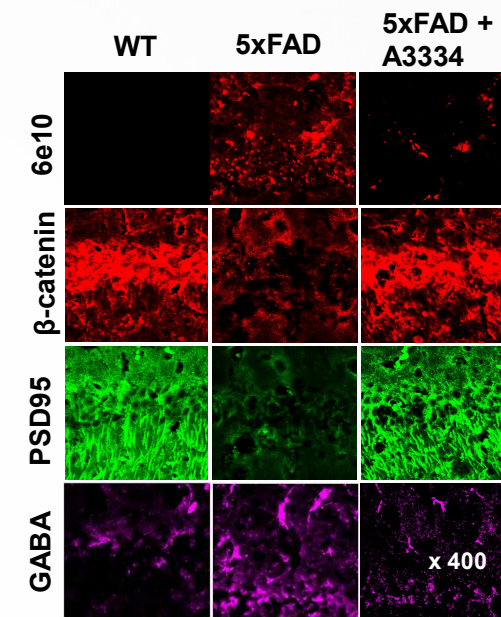
2) Genetic confirmation



2

5xFAD AD model mouse did not reveal AD phenotypes when that was hybrid with a $Cxxc5^{-/-}$ mouse in Hippocampus tissues.

3) Pharmacological replication



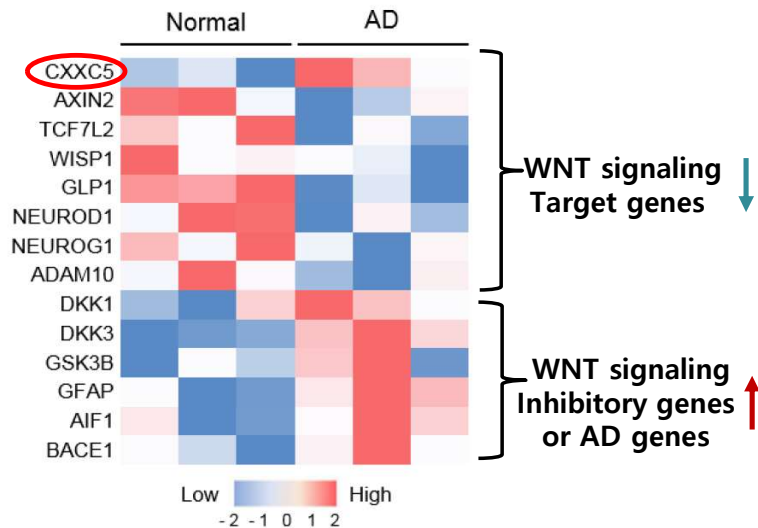
3

CXXC5-Dvl PPI inhibitor suppressed the pathological phenotypes of 5xFAD AD model mouse in Hippocampus tissue

Regenerative WNT pathway is suppressed in AD patient tissue cells

[Makes AD is difficult be cured]

The regenerative “Wnt/ β -catenin signaling is suppressed in AD patient tissues



Genes involving neural stem cell activation & neurogenesis

- Neurogenin-1 (NEUROG1)
- Neurogenin-2 (NEUROG2)
- NeuroD1

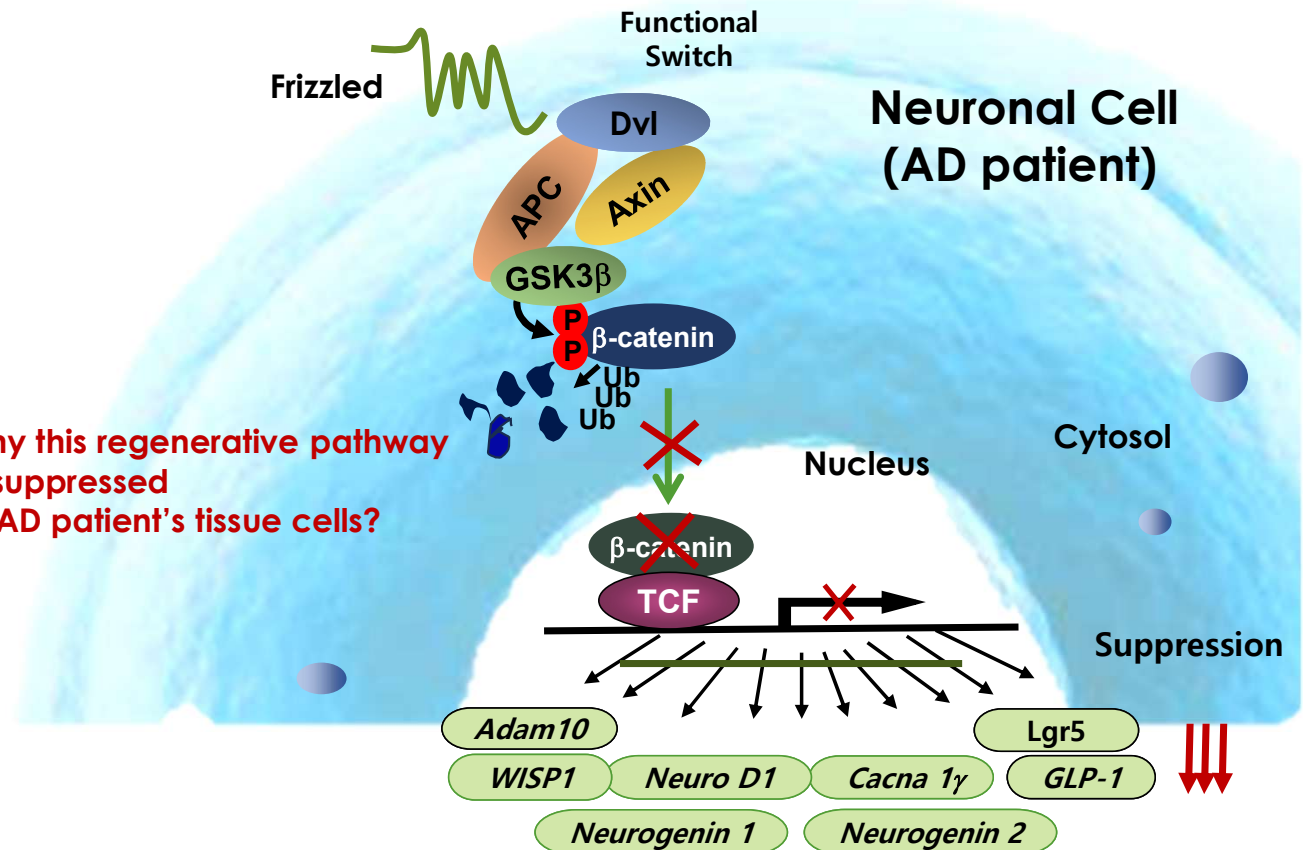
Stemness & regeneration

- LGR5, Axin2, NeuroD1

Why this regenerative pathway is suppressed in AD patient's tissue cells?

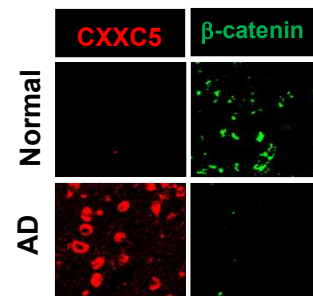
- ❖ The WNT pathway target genes are tightly related with neural function, regeneration of cells, and their expression is suppressed in AD patient tissue cells.

Restoring Wnt/ β -catenin signaling is a promising therapeutic strategy for Alzheimer's disease
Lin et al. Molecular Brain 12, 104 (2019)



AD pathogenesis and molecular target suggestion

[Hippocampus]



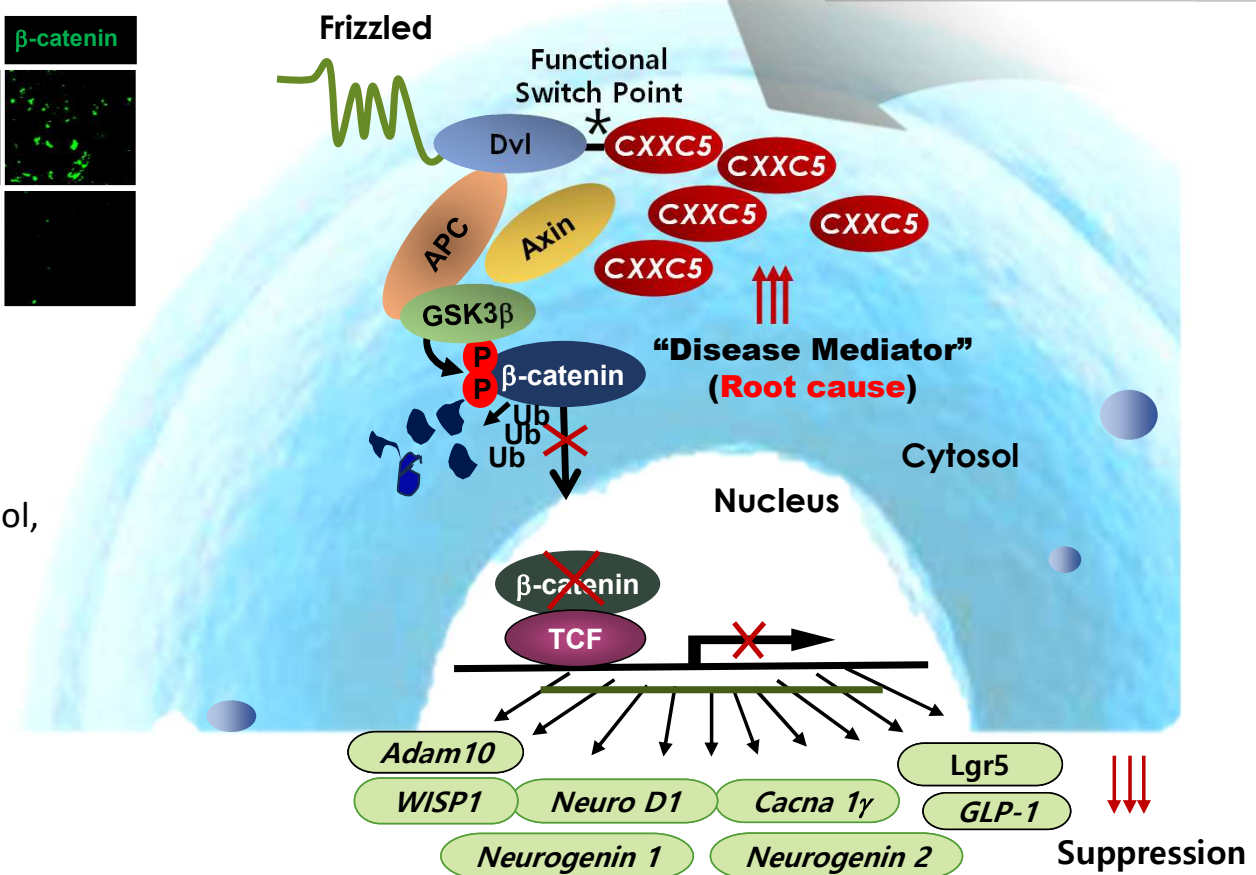
[Pathogenesis]

CXXC5 is a disease mediating factor; A Root Cause

In AD patient tissue cells, CXXC5 is increased in their cytosol, binds to an upstream component “Dishevelled (Dvl)” and subsequently inactivates the WNT pathway via β -catenin degradation, and suppresses expression of target genes involving regeneration and neural functions.

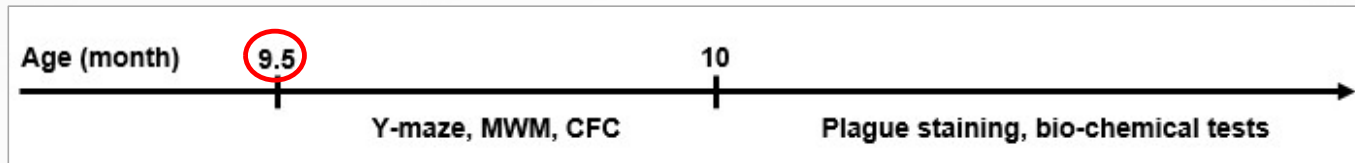
AD-Neuronal Cell

Aging
Genetics (gene mutations)
APP, MAPT, and PSEN1 etc.

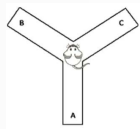


■ *Cxxc5* KO mice resistant to the pathological phenotypes of Alzheimer's

[Indicates CXXC5 is a key mediator for Alzheimer's disease]

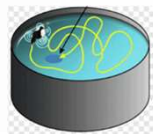


{Behavior Tests}



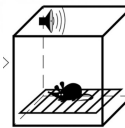
[Y Maze]

Exploring new environments



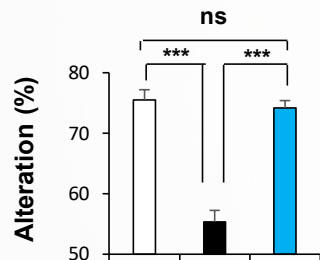
[Morris Water Maze]

Spatial learning/ short memory

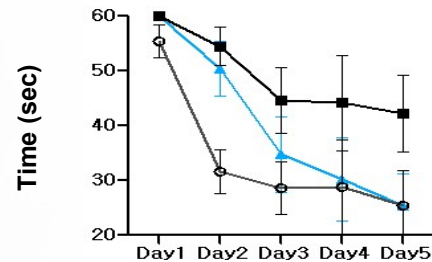


[Contextual Fear Conditioning]

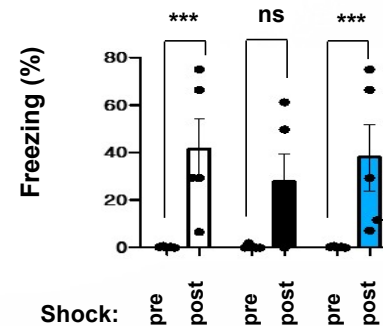
Learn and remember



□ WT
■ 5xFAD
■ *Cxxc5*^{-/-}/5xFAD

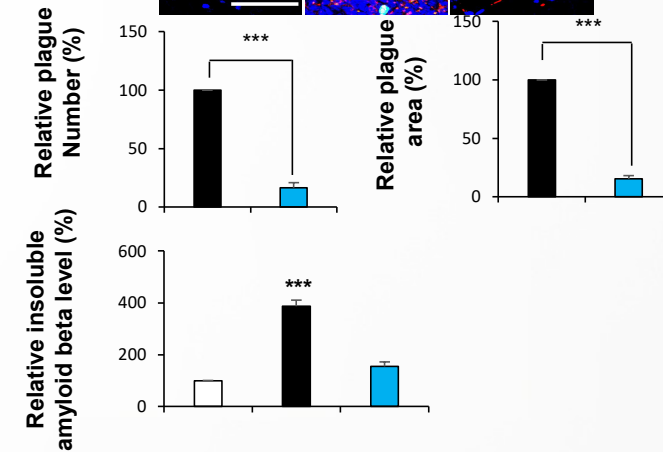
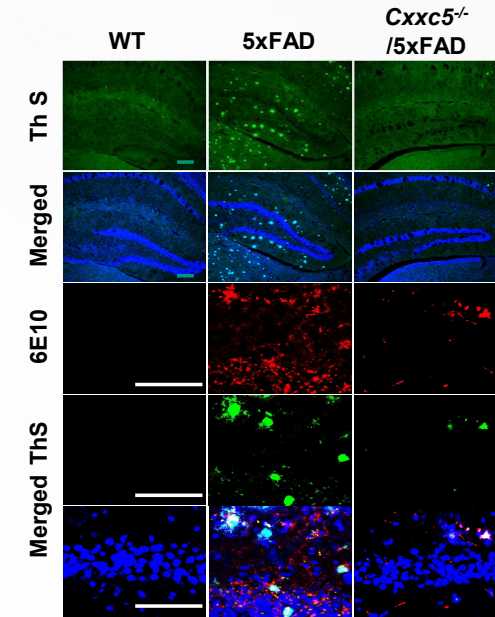


○ WT
■ 5xFAD
▲ *Cxxc5*^{-/-}/5xFAD



Shock: pre post pre post pre post

□ WT
■ 5xFAD
■ *Cxxc5*^{-/-}/5xFAD



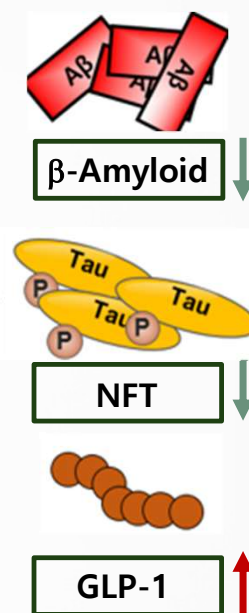
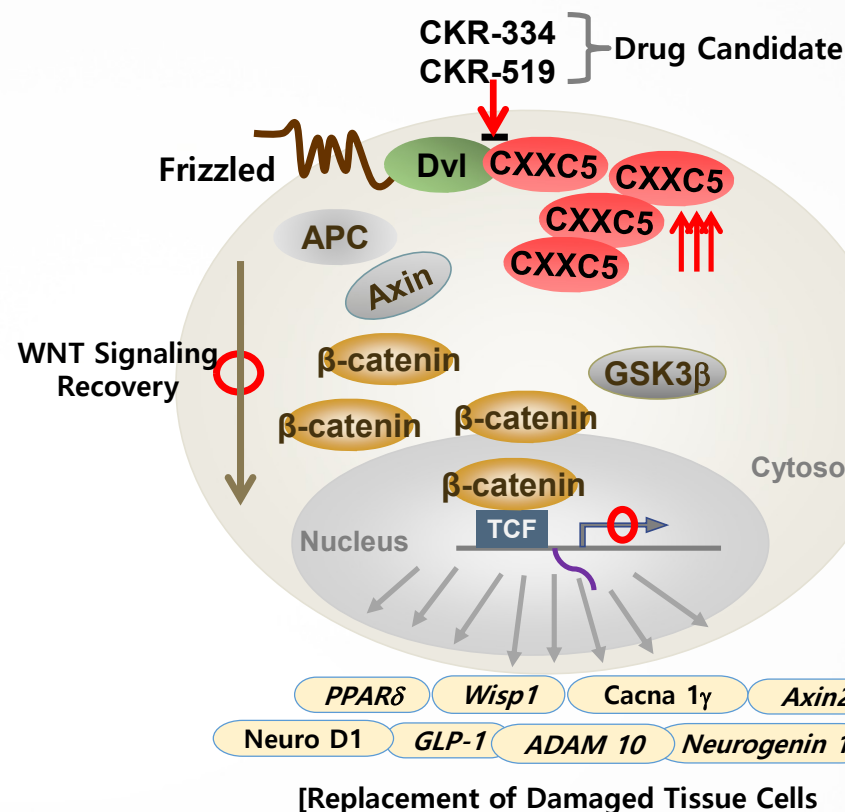
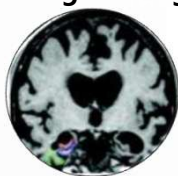
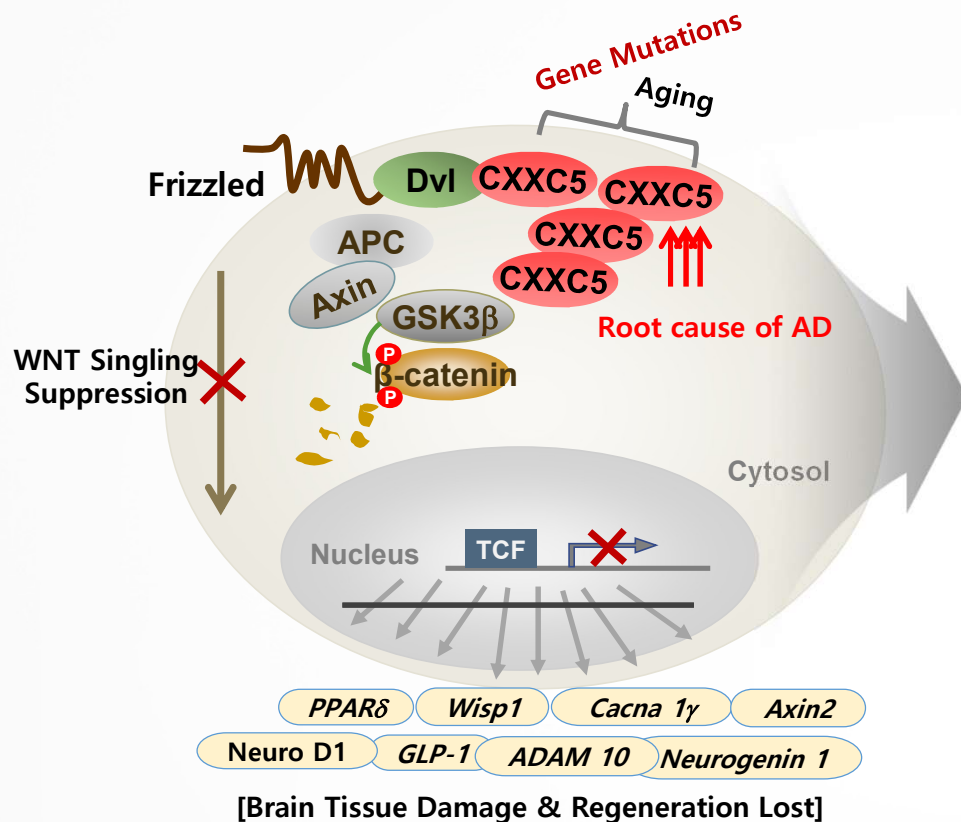
Targeting Alzheimer's Disease by Specific Cytosolic Functional Inhibition of CXXC5



① WNT Signaling Inactivation by Cytosolic CXXC5

② WNT Signaling Recovery by CXXC5 Function Inhibition

③ Improvement of the Major Disease Phenotypes



Brief Historical Overview To Obtain Drug Candidates; [Small molecule CXXC5-Dvl PPI inhibitors]

Screening & selection of small molecules restore WNT signaling by CXXC5-Dvl PPI inhibition

1. Discovery of “PTD-DBM peptide” specifically controlling cytosolic function (WNT signaling suppression) of CXXC5

Developed a novel cell-penetrating peptide that specifically blocks the CXXC5–Dishevelled (Dvl) PPI and subsequently restores suppressed WNT/ β -catenin signaling

2. Establishment of a novel screening platform monitoring the CXXC5-Dishvelled PPI-Platform

By using the PTD-DBM peptide, an innovative screening system to identify small molecules that specifically disrupt the CXXC5–Dvl interaction was established.

3. Primary small-molecule screening; CXXC5-Dvl PPI Inhibitors

Screened small molecule libraries containing approximately 8,000 compound, therefore, identified initial hit compounds through multiple analytical approaches.

4. WNT signaling activator screening

Selected compounds that restore WNT/ β -catenin signaling using cells harboring pTPOPLASH-WNT signaling reporter & further analyzed their functions with pharmacological analyses.

5. Lead optimization

Synthesized and evaluated over 300 derivatives of lead compounds through repeated cycles of MOA confirmation, in vitro testing, and in vivo efficacy studies for multiple metabolic studies.

6. Drug candidate nomination

Advanced optimized compounds into drug development as regenerative small-molecule candidates.



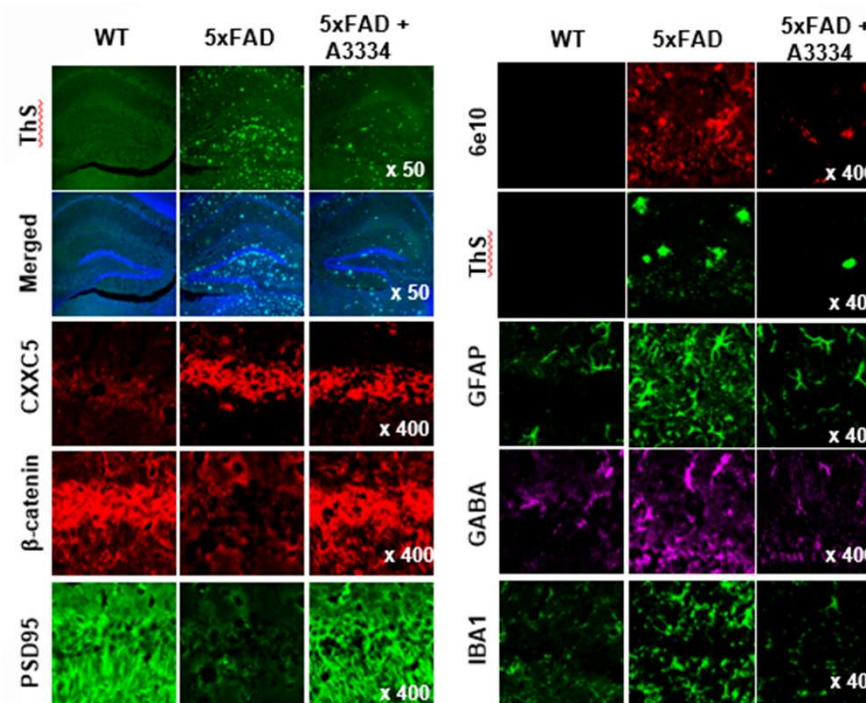
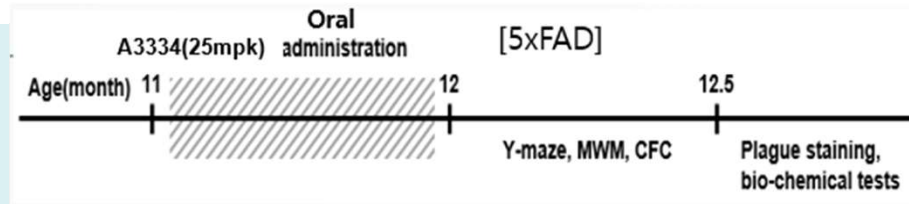
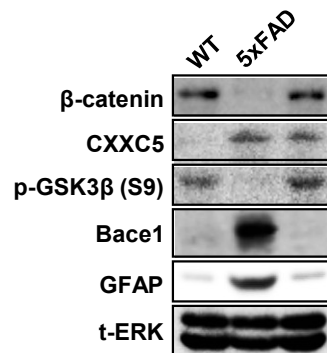
Pharmacologic Replication

Disease-Modifying Therapeutic Effects by CKR-334 (A3334)

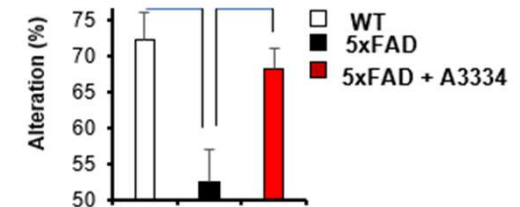


Therapeutic effects of CKR-334, a small molecule inhibitor interfering the CXXC5-Dishvelled was tested by using age-induced AD developed 5xFAD mice.

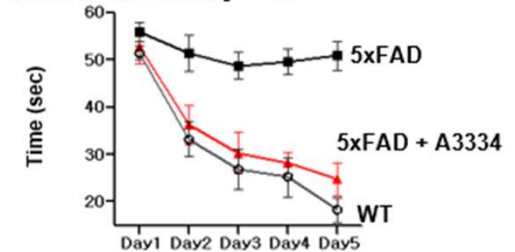
Both MOA and efficacies were monitored by immunohistochemical analyses of hippocampus tissues of 12-month-old mice; WT, 5xFAD, or 5xFAD mice orally administered A3334 for 1 month before scarification of mice.



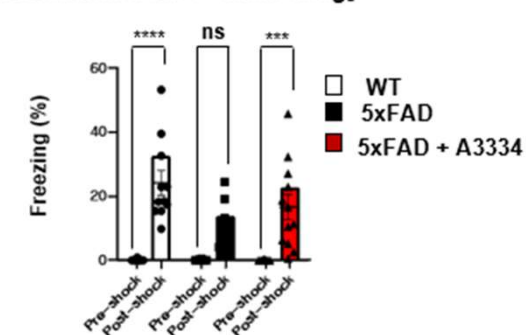
[Y Maze]



[Morris Water Maze]



[Contextual Fear Conditioning]



CKR-519

Designed to overcome the solubility limitations of indirubin-family compounds
while securing matter patentability

Novel structural modification; Breeding planarity, introduction of a solubilizing group/bioisosteric modification-Tailored heterocycle incorporation to optimize core properties.

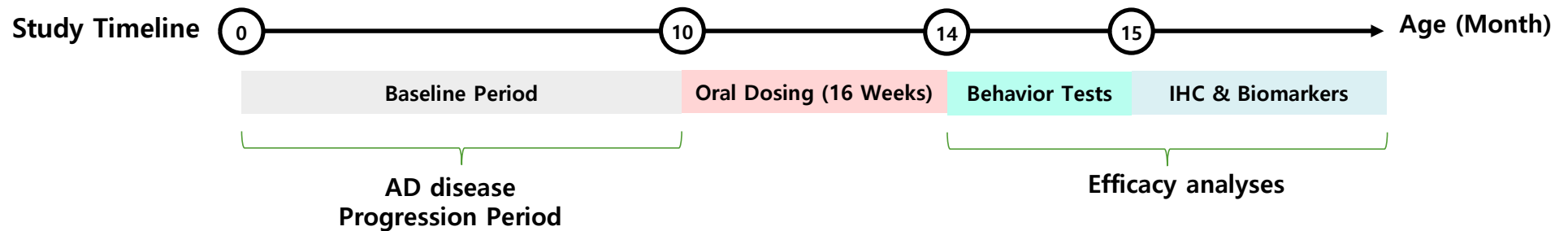
1) Solubility Leap, 2) Enhance Druggability, 3) Novelty for Patents

- ✓ Improvement of solubility-aqueous solubility
 - ✓ Enhanced cellular permeability with functional WNT signaling activation via CXXC5–Dvl PPI inhibition (by comparing *in vitro* CXXC5-Dvl PPI *vs* cellular functionality cf. pTOFOPLASH reporter activity)
 - ✓ Improvement of oral bioavailability by pure structural change without formulation.
 - ✓ Improved physicochemical properties support further in vivo PK, absorption, and efficacy validation.
- ◆ In vivo efficacies by oral administration was confirmed for the multiple target diseases

In vivo efficacy studies for CKR-519



Alzheimer's Disease-5xFAD model mouse



- 5xFAD mice were maintained until 10 months of age, followed by 16 weeks of daily oral administration.
- After 16 weeks of oral administration, behavioral tests were conducted for 4 weeks while treatment was continued.
- Behavior test; Y-maze, Barnes maze, Novel object recognition test etc.

Results Description

- ✓ Efficacy was observed, particularly at low concentrations, in behavioral analyses using the Y-maze test, including alternation ratio and total entries, Barnes maze test for spatial learning and memory, and novel object recognition test.
- ✓ Histological findings correlated with the observed behavioral improvements, based on the assessment of multiple markers, including β -amyloid pathology markers 6E10 and ThS, GFAP, Iba1, NeuN, MAP2, and β -catenin, in hippocampal and cortical tissues.

Key Features of CXXC5-Dvl PPI Inhibitors



❖ Novel Targeting Approach; CXXC5-Dishvelled PPI

Our first-in-class, orally administered small molecule targets the CXXC5–Dishevelled (Dvl) protein–protein interaction (PPI) by selectively disrupting the binding of cytosolic CXXC5 to the upstream Wnt pathway mediator Dvl. This restores Wnt/ β -catenin signaling that is otherwise pathologically suppressed in Alzheimer’s Disease (AD). The therapeutic relevance of targeting cytosolic CXXC5 through CXXC5–Dvl PPI inhibition is strongly supported by the observation that overexpression of cytosolic CXXC5 in the hippocampal and prefrontal cortex neurons of AD patients and 5xFAD model mice correlates with marked suppression of Wnt signaling. This was further validated genetically: crossing 5xFAD mice with *Cxxc5*^{-/-} mice completely prevented the pathological phenotypes of the 5xFAD model, including synaptic dysfunction and cognitive deficits. These findings establish cytosolic CXXC5 as a critical disease-driving factor and confirm the therapeutic potential of CXXC5–Dvl PPI inhibitors for Alzheimer’s Disease.

❖ Long-Lasting Therapeutic Effect

- The WNT pathway is closely associated with AD, particularly its suppression, and restoration has been suggested by others as a promising therapeutic strategy. Key pathological features of AD—including β -amyloid accumulation, tau phosphorylation, and GLP-1 suppression—are strongly linked to WNT signaling inactivation.
- Therefore, our approach, which restores suppressed WNT signaling by blocking the function of a disease-driving negative regulator, represents a highly attractive therapeutic strategy.
- Through repeated administration into the mouse developed AD phenotypes, the candidate demonstrates therapeutic effects by regenerating and replacing damaged tissue cells, thereby restoring multiple pathological features of AD—unlike other available or investigational therapies

❖ Oral Delivery

- Oral formulation.

❖ Safe Wnt Pathway Targeting

- Our approach restores suppressed Wnt/ β -catenin signaling rather than pathologic overstimulating it, which is often caused by mutations such as *APC* loss or *CTNNB1* (β -catenin) activation, provide a safe approach targeting WNT pathway in multiple diseases suppressed WNT signaling by CXXC5..

❖ Disease-Tissue Specific Action

- The drug candidate specifically targets diseased cells with high cytosolic CXXC5 expression while sparing normal tissue cells.
- Administration to healthy mice revealed no significant abnormalities in any organs or tissues.